

# MAX YUAN

647-819-0765 | maxyuan081205@gmail.com | Portfolio: [notmax08.github.io](https://notmax08.github.io) | [github.com/NotMax08](https://github.com/NotMax08) |

## TECHNICAL SKILLS

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**Languages:** Python, C++, Java

**Environments:** Ubuntu, Linux CLI, Git

**Simulation & Vision:** NVIDIA Isaac Sim, Isaac Lab, OpenCV, YOLOv8

**Robotics & Control:** ROS2, Rviz, Nav2, slam\_toolbox, AMCL, PID Control, Inverse Kinematics, Odometry, Onshape

**Machine Learning:** Reinforcement Learning (PPO), Imitation Learning, SFT, VLA ( $\pi_0$ ), rsl\_rl, pytorch

## EDUCATION

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**University of Waterloo | Sept 2026 – Apr 2031**

*Bachelor of Applied Science, Mechatronics Engineering, Honours Co-op | Waterloo, ON*

## PROJECTS

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[The Brackey Way](#) | *Autonomous VLA Sandwich Assembly on Bracket Bot* | **Sept 2026**

**1st Place**, BOTS Hackathon (WAT.ai vs. UTMIST, Battle of the Schools)

- Integrated a  $\pi_0$  Vision-Language-Action foundation model via supervised fine-tuning (SFT), collecting 100+ teleoperated input-output pairs (video footage and text prompt as input, joint position and velocity as output)
- Reduced overfitting by varying object positioning and lighting across demonstration episodes by 30%
- Localized ingredients with HSV threshold segmentation, applying inverse kinematics for manipulation accuracy
- Built a finite state machine switching between classical (CV + IK) and learning-based (VLA) controllers

[Custom Quadruped Locomotion](#) | *Unitree Go2 PPO in Isaac Lab* | **Aug 2026 – Sept 2026**

Solo Project: Custom ManagerBasedRLEnv for flat-terrain Go2 locomotion, trained with PPO via NVIDIA Isaac Lab

- Built a custom ManagerBasedRLEnv for Go2 flat-terrain locomotion, defining the observation space, 12-DOF joint-position action space, and reward terms for a PPO policy trained in Isaac Lab
- Designed a sigmoid curriculum that ramps the foot-lift reward over training, forcing the policy to establish stable balance before the reward begins shaping gait quality

[Autonomous Maze Navigation](#) | *Nova Carter in Isaac Sim* | **May 2026 – July 2026**

Solo Project: Full ROS2 Nav2 stack w/ NVIDIA Isaac Sim Ubuntu 24.04

- Integrated SLAM (slam\_toolbox), AMCL localization, Nav2 with a Pure Pursuit controller, and Rviz for debugging, reaching goals consistently across repeated runs
- Constructed the full TF tree (map → odom → chassis\_link → sensors) via Isaac Sim action graphs, debugging QoS (Quality Of Service) mismatches, sensor rate bottlenecks, and frame authority conflicts

## EXPERIENCE

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WATonomous | Humanoid Sub Team | Sept 2026 - Present

- Contributing to the RL & Autonomy stack on the humanoid sub team, with upcoming work in reinforcement learning (Isaac Lab) and simulation-based digital twins for  $\pi_0$  VLA model development

[Co-Captain, Lead Programmer, Driver | FIRST Tech Challenge Team 18844](#) | **Sept 2023 - April 2025**

**1st Place Connect Award @ 2025 FIRST World Championships as Team Canada Representatives**

- Distributed work across the programming subdivision, scoping tasks by skill level and integrating contributions
- Ran robot showcases and hands-on SPIKE Robotics workshops inspiring & teaching 60+ children across the GTA
- Built a PID pathing system fusing 2-wheel odometry with IMU data to localize the robot in global coordinates
- Automated a pickup system with using OpenCV and a custom-trained CNN to estimate distance and orientation